

Network Monitoring — Guided Notes ANSWER KEY

Unit 3.1.2 | CompTIA Network+ N10-009 Obj. 3.1 | N10-008 3.1

For instructor use / distribute after students complete the notes.

Question 1

SNMP (Simple Network Management Protocol) uses a _____ model where the monitoring server polls devices, and a _____ model where devices send unsolicited alerts when a threshold is crossed. The alert message is called a _____.

Answer: Poll/pull; trap (push); trap (SNMP trap).

Question 2

SNMPv1 and v2c transmit community strings in _____. SNMPv3 adds _____ and _____. Which version should be used on any production network and why?

Answer: Cleartext (plaintext); authentication; encryption. SNMPv3 — sending community strings in cleartext exposes management credentials to anyone capturing traffic on the network.

Question 3

NetFlow exports _____ data from routers and switches — source/destination IP, port, protocol, byte count, and timestamps — without capturing the actual payload. Name two security use cases for NetFlow data.

Answer: Traffic flow (metadata); detecting port scans (one host hitting many ports in rapid succession), identifying data exfiltration (large unexpected outbound transfers), spotting C2 traffic patterns, finding rogue DHCP servers.

Question 4

Syslog collects log messages from network devices to a central _____ server. Severity levels range from _____ (most critical, level 0) to _____ (least critical, level 7). Why is centralized logging more valuable than reading logs on each device individually?

Answer: Syslog; Emergency; Debug. Centralized logs allow correlation across devices — an attack that touches five switches appears as a pattern in one place rather than as isolated events on five separate devices.

Question 5

Interface utilization monitoring tracks the percentage of available bandwidth in use on each link. A link consistently running above _____ % utilization is considered congested and a candidate for upgrade or load balancing. What user symptom typically accompanies sustained high utilization?

Answer: 70–80% (accept any reasonable threshold, typically 70-80%); slow file transfers, buffering video, high latency, dropped VoIP calls.

Question 6

A network monitoring system uses _____ to determine if a device is reachable. If a device stops responding, the system generates an _____ to notify the administrator. What is the risk of alert fatigue in a monitoring system?

Answer: ICMP ping (or SNMP polling); alert/alarm. Alert fatigue occurs when too many low-priority alerts are generated — administrators begin ignoring or dismissing alerts automatically, causing real critical events to be missed.

Question 7

Port mirroring (SPAN) copies traffic from one or more _____ to a designated _____ port connected to a monitoring device like Wireshark or an IDS sensor. What is the limitation of using a SPAN port for IDS compared to an inline IPS?

Answer: Source ports (or VLANs); destination (monitor). A SPAN-connected IDS can only detect and alert — it receives a copy of the traffic and cannot block anything because it is not in the data path.

Question 8 — Real World Application

REAL WORLD: A school's internet link was saturated every day from 2–3 PM for three weeks before anyone noticed. No monitoring was in place. Describe a monitoring setup using two tools from this lesson that would have detected this within the first day, and what each tool would have shown.

Answer: 1. Interface utilization monitoring (SNMP polling the uplink interface) — would have shown the uplink hitting 95-100% utilization at 2 PM daily, triggering a threshold alert immediately. 2. NetFlow analysis — would have identified which internal IPs and which protocols (likely video streaming or large file transfers) were consuming the bandwidth, enabling a targeted fix.